

Notes: Diamond and Rajan (JPE, 2001)

Liquidity Risk, Liquidity Creation, and Financial Fragility: A Theory of Banking

Contents

1	Big picture	3
2	Model setup and key objects	3
2.1	Projects, timing, and cash flows	3
2.2	Financing contracts	4
2.3	Relationship lending and liquidation values	4
2.4	Limited commitment and renegotiation	4
2.5	Liquidity shock	4
2.6	Assumptions and their meaning	5
2.7	Illiquidity premium	5
3	Models and theoretical results (all equations + interpretation)	5
3.1	The entrepreneur's objective	5
3.2	Renegotiation and maximum extractable repayments	6
3.3	Lemma 1: optimal direct lending outcomes when the liquidity shock is contractible	6
3.4	Private information about liquidity needs: Corollary 1	7
3.5	Corollary 2: the source of illiquidity	7
3.6	Why the resale price is low: the two cases	7
3.7	Example and comparative statics	8
3.8	Financial intermediation with demand deposits	8
3.9	Why any attempt to underpay depositors leads to a run	9
3.10	Lemma 2, Lemma 3, and Proposition 1	9
4	Mechanism, intuition, and policy implications (what makes the result work)	10
4.1	What ingredients are essential	10
4.2	Why fragility is useful rather than wasteful	10

4.3	Distinguishing commercial banks from other intermediaries	10
4.4	Narrow banking, bank capital, deposit insurance, and suspension	10
4.5	Relation to the literature	11
4.6	Limitations and scope	11
5	Tables and figures	11
5.1	Figures 1 and 2: bargaining trees at dates 2 and 1	11
5.2	Table 1: timeline	12
5.3	Lemma 1 and the example: Figures 3 and 4	12
5.4	Figure 5: depositor response to a low offer	13
6	Short summary	13
7	Conclusion	14
7.1	Core conclusion (what the paper establishes)	14
7.2	Economic intuition	14
7.3	Takeaway	14

1 Big picture

- **Question.** Why do banks finance illiquid loans with fragile demand deposits? Is bank fragility merely a flaw, or can it be a useful institutional feature?
- **Setting.** The paper studies a two-period lending relationship between an entrepreneur and a relationship lender who develops special loan-collection or liquidation skills. The lender may face an interim liquidity shock before the loan matures.
- **Core challenge.** Relationship-specific skills make direct lending feasible, because the lender can extract repayment better than outsiders. But these same specific skills also make the loan *illiquid*: if the lender needs money before maturity, she cannot sell or pledge the loan for its full value.
- **Main theoretical result.** A bank with a fragile capital structure — in particular, demand deposits subject to runs — can commit to pass through to depositors the full value it can extract from borrowers. This commitment makes otherwise illiquid loans financeable without a liquidity premium.
- **Key mechanism.** If the banker ever tries to renegotiate deposit claims downward, depositors run. A run disintermediates the banker, transfers the loan to unskilled outsiders, and destroys the banker's rents. Anticipating this, the banker honors deposits and so can borrow against the loan's full collectible value.
- **Liquidity creation.** Fragility creates liquidity on both sides of the balance sheet:
 - borrowers are protected from the lender's own liquidity needs, and
 - depositors obtain liquid claims even though the underlying loans are illiquid.
- **Policy implication.** Stabilization policies such as narrow banking, high capital requirements, complete deposit insurance, or suspension of convertibility may reduce the bank's ability to create liquidity.
- **Economic takeaway.** Bank fragility is not just a source of instability. In this paper, fragility is the commitment device that makes liquidity creation possible.

2 Model setup and key objects

2.1 Projects, timing, and cash flows

- The economy lasts for two periods and three dates: date 0, date 1, and date 2.
- An entrepreneur has a project that requires investment of up to 1 at date 0.
- If the entrepreneur works on the project, it generates deterministic cash flows C_t at date t .
- There is also a storage technology that returns 1 for every dollar invested.
- The entrepreneur has no funds initially and must borrow from outsiders.

2.2 Financing contracts

- A financing contract is summarized by required repayments P_t at different dates.
- If the entrepreneur defaults, the lender gets control rights over the asset and may liquidate it.
- The contract space is broad: debt with finite promised payments is included, and full lender control is a limiting case obtained by setting promised payments high enough.

2.3 Relationship lending and liquidation values

- The initial lender is a **relationship lender**. Because she learns the entrepreneur's business, she can liquidate the project better than outsiders.
- If she takes control before date t , she can put the asset to alternative use and obtain liquidation value X_t .
- Outside lenders who come later can generate only βX_t , where

$$0 \leq \beta < 1.$$

In the paper, this captures the idea that outsiders lack relationship-specific collection skills.

- These skills are not simply information in the abstract; they are a source of bargaining power over future surplus.

2.4 Limited commitment and renegotiation

- The entrepreneur cannot commit his human capital beyond the current date. He can always threaten to walk away before future cash flows are produced.
- The relationship lender also cannot commit to using her specific skills on behalf of others in the future.
- As a result, loan contracts can be renegotiated.
- For simplicity, the entrepreneur has all the bargaining power: after default, he can make a take-it-or-leave-it renegotiation offer.

2.5 Liquidity shock

- With probability v at date 1, the relationship lender receives a liquidity shock and becomes impatient.
- If shocked, she values one unit of date 1 consumption at R units of date 2 consumption, where

$$R > 1.$$

- The shock is privately observed by the lender.
- The model assumes there is no aggregate liquidity shortage: enough patient investors exist that storage is always available at the margin.

2.6 Assumptions and their meaning

- **Assumption 1.**

$$\min\left\{C_1 + C_2, \frac{C_1 + C_2}{X_1}\right\} > R > 1.$$

The project is socially worth continuing even when the lender is impatient.

- **Assumption 2.**

$$C_2 > X_2.$$

At date 2, continuation dominates liquidation.

- **Assumption 3.**

$$\max\{X_1, X_2\} \geq 1.$$

Without liquidity shocks, the entrepreneur can pledge enough to get the project financed.

2.7 Illiquidity premium

The paper defines the **illiquidity premium** as the extra expected payment the entrepreneur must make over and above the expected payment the lender would obtain from storing her funds. The formula is

$$\text{Illiquidity premium} = v(R - 1)(1 - \text{date-1 cash flow to impatient lender}).$$

Interpretation. A loan is illiquid when it pays too little precisely in the state when the lender most values cash, namely when she is hit by the liquidity shock at date 1. Even a loan that is riskless if held to maturity can therefore be risky for a lender with liquidity needs.

3 Models and theoretical results (all equations + interpretation)

3.1 The entrepreneur's objective

- The entrepreneur wants to maximize expected net payoff subject to the lender's individual rationality.
- Since the lender's outside option is storage, the entrepreneur must promise the lender expected utility at least

$$(1 - v) + vR.$$

- The entrepreneur's priorities are:

1. minimize the probability of liquidation, and then
2. minimize the illiquidity premium by delivering as much cash as possible when the lender is impatient.

3.2 Renegotiation and maximum extractable repayments

At date 2, if the original loan is still in the hands of the relationship lender, the entrepreneur renegotiates any promised payment above X_2 down to X_2 . Hence actual payment at date 2 is

$$\min\{P_2, X_2\}.$$

At date 1, the key distinction is whether the lender is patient or impatient.

- If the lender is **patient**, she can either liquidate immediately for X_1 or continue and ultimately extract X_2 . So the maximum total enforceable payment is

$$E^{\sim I} = \max\{X_1, X_2\}.$$

- If the lender is **impatient**, she can:
 - liquidate immediately for X_1 ,
 - sell the loan at date 1 for S ,
 - or keep the loan and wait, yielding present value X_2/R .

So the maximum enforceable present value is

$$E^I = \max\left\{X_1, S, \frac{X_2}{R}\right\}.$$

- If the project is not liquidated and the loan is sold, the maximum immediately available cash is

$$C_1 + S.$$

Interpretation. These expressions summarize the whole problem. The lender's special skills let her extract more than outsiders, but if she needs liquidity before maturity, she may be forced to realize only the resale value S , not the full continuation value X_2 .

3.3 Lemma 1: optimal direct lending outcomes when the liquidity shock is contractible

Suppose first that the entrepreneur can write a contract directly contingent on whether the lender is impatient or not.

- If

$$\min\{C_1 + S, E^I\} > 1,$$

the entrepreneur can finance the project and even provide **liquidity insurance**: the loan can have a *negative* illiquidity premium.

- If

$$\min\{C_1 + S, E^I\} = 1,$$

the loan is exactly **liquid**: it has zero illiquidity premium.

- If

$$\min\{C_1 + S, E^I\} < 1,$$

the entrepreneur may still be able to finance the project, but now he must pay a **positive illiquidity premium** or allow liquidation when the lender is impatient.

- If the lender's individual rationality cannot be met even with such contract terms, the project is not financed.

Interpretation. Even with fully state-contingent contracts, direct relationship lending can be illiquid. The entrepreneur may need to compensate the lender for the fact that the loan performs poorly exactly in the lender's high-marginal-value state.

3.4 Private information about liquidity needs: Corollary 1

In reality, the lender's liquidity shock is privately observed, so the entrepreneur cannot directly write a shock-contingent contract. He can only offer a menu from which the lender self-selects.

A key incentive constraint is that the patient type must weakly prefer the patient schedule in market-value terms, which implies that the impatient type cannot be given strictly better liquidity insurance than is incentive compatible. The paper shows:

- a liquid loan is still possible when the conditions for zero illiquidity premium are satisfied;
- but **liquidity insurance is impossible**: a loan cannot be “more than liquid” once liquidity needs are private information;
- relative to the full-information case, the loan is weakly more illiquid and is sold in weakly fewer states;
- liquidation does not increase solely because of this informational friction.

Interpretation. Private information prevents the entrepreneur from perfectly tailoring payments to the lender's need for liquidity. It blocks negative illiquidity premia, but not necessarily liquidity itself.

3.5 Corollary 2: the source of illiquidity

If the loan can be sold for the full value that the relationship lender can extract at date 2, namely if

$$S = X_2,$$

then the optimal loan is always liquid.

Interpretation. Illiquidity is not fundamentally about uncertainty in project cash flows. It comes from the wedge between what the relationship lender can extract if she keeps the loan and what outsiders will pay for it if she must raise cash early.

3.6 Why the resale price is low: the two cases

The paper studies two cases.

- **Case 1: sale destroys the lender's skills.** If selling the loan or taking the outside opportunity destroys the relationship lender's collection ability, then a buyer can collect only

$$S = \beta X_2.$$

- **Case 2: the lender keeps her skills after sale.** Even if the relationship lender retains her skills, the loan buyer still owns the control rights and can negotiate directly with the entrepreneur. Because the lender cannot commit to using her skills on behalf of the buyer without trying to capture rents, the buyer again pays only

$$S = \beta X_2.$$

Interpretation. Specific human capital is necessary for illiquidity, but not sufficient. Illiquidity also requires *noncommitability*: the relationship lender cannot credibly promise to deploy her skills for the benefit of outsiders.

3.7 Example and comparative statics

The paper illustrates the logic with

$$X_1 = 0.9, \quad X_2 = 1.1, \quad C_1 = 0, \quad C_2 = 1.5, \quad R = 1.4, \quad S = 0.8.$$

- Since $\max\{X_1, X_2\} = 1.1$, the entrepreneur can never commit to paying more than 1.1, even though the project generates 1.5 in total.
- For low liquidity-shock probability v , the lender sells the loan when shocked and the entrepreneur finances the project using a standard long-term debt contract.
- Once v rises above about 0.26, the entrepreneur can no longer satisfy the lender purely through resale proceeds and must allow liquidation when the lender is hit by the shock.
- Once v rises above about 0.42, even that is not enough and the entrepreneur cannot borrow at all.

Interpretation. Greater lender liquidity risk can make otherwise productive loans effectively unavailable. This is the direct-lending problem that banking solves.

3.8 Financial intermediation with demand deposits

Now suppose the relationship lender becomes a **banker** and, at date 1, borrows against the project loan by issuing demand deposits to many small unskilled depositors.

- Each depositor is promised a fixed amount per unit deposited.
- If the banker fails to pay on demand, depositors can seize financial assets with market value equal to their promised payment.
- Deposit claims are served on a **first-come, first-served** basis.

The benchmark contract that creates liquidity promises

$$d_1 = d_2 = 1.$$

That is, depositors can withdraw one unit at date 1 or leave funds until date 2 and still receive one unit.

Interpretation. This exactly matches the return and liquidity of storage. Deposits become liquid, even though the underlying project loan is not.

3.9 Why any attempt to underpay depositors leads to a run

Suppose the banker tries to renegotiate deposit payments downward at date 2 and offers less than full value. If a fraction f of depositors run, the expected payoff from running is

$$\bar{r}(f) = \begin{cases} d_2 & \text{if } f \leq \frac{\beta X_2}{d_2}, \\ \frac{\beta X_2}{f} & \text{if } f > \frac{\beta X_2}{d_2}. \end{cases}$$

- If enough assets remain, a runner gets paid in full.
- If too many depositors run, assets are rationed, but each depositor still wants to run because those who accept the low offer are paid only from the residual assets left after runners are served.

Interpretation. The first-come-first-served rule creates a collective-action problem among depositors. Even if all depositors would be better off coordinating, each one individually wants to run first. This is exactly the discipline device the banker needs.

3.10 Lemma 2, Lemma 3, and Proposition 1

- **Lemma 2.** If there are sufficiently many depositors, any attempt by the banker to offer less than full value leads to disintermediation and drives the banker's rents to zero. Therefore, the banker can commit to pass through everything she can collect.
- **Lemma 3.** If the entrepreneur borrows directly from many unskilled lenders instead of from a banker, he can still promise at most

$$\beta X_2.$$

So multiple short-term creditors alone do *not* solve illiquidity.

- **Proposition 1.** A bank that issues demand deposits to sufficiently many depositors, with equal promised payments at dates 1 and 2 and first-come-first-served withdrawal rights, can commit the banker to collecting loans on behalf of depositors. Deposits are liquid even though the underlying loans are illiquid.

Interpretation. The special combination of *relationship lending* and *fragile deposit finance* is what creates bank liquidity. Neither ingredient by itself is enough.

4 Mechanism, intuition, and policy implications (what makes the result work)

4.1 What ingredients are essential

- **Specific skills.** The relationship lender must be able to collect more than outsiders.
- **Noncommitability.** She must be unable to commit those skills costlessly on behalf of others.
- **Fragile liabilities.** Deposits must be demandable and subject to first-come-first-served withdrawal.
- **Many depositors.** A large number of claimholders makes a unilateral run effective in destroying the banker's rents.

Without these ingredients, the banking result disappears.

4.2 Why fragility is useful rather than wasteful

- In the model, fragility is valuable because it commits the banker not to renegotiate with depositors.
- The threat of a run is enough; actual runs need not occur in equilibrium when the banker honors deposits.
- Depositors do not need sophisticated private monitoring. They only need the ability to punish harshly once trouble becomes visible.

Interpretation. Fragility is a commitment technology, not merely a source of panic. That is the paper's central conceptual contribution.

4.3 Distinguishing commercial banks from other intermediaries

- **Money market mutual funds.** These are run-proof because depositors own a pro rata share of marked-to-market assets. Since their assets are already liquid, the fund does not create liquidity; it merely passes through underlying liquidity.
- **Insurance companies.** Insurance contracts pay out conditional on observable and verifiable events. By contrast, banks provide liquidity against needs that are often private or unverifiable.
- **Investment banks and venture capitalists.** Their value often lies in future deal making or ongoing managerial input, not merely in extracting repayment from existing positions. Because they continue to create new surplus after financing, a run would not discipline them in the same way.

4.4 Narrow banking, bank capital, deposit insurance, and suspension

- **Narrow banking.** Forcing demand deposits to be backed only by liquid assets would eliminate the commitment role of deposits and thus reduce bank liquidity creation.

- **Bank capital.** In the paper's certainty environment, fragility is useful. With aggregate uncertainty, some capital may be desirable as a buffer, because it trades off commitment against resilience.
- **Complete deposit insurance.** If all deposits are fully insured, depositors no longer run and so no longer discipline the banker. The bank loses the ability to create liquidity through fragility.
- **Suspension of convertibility.** If a bank can suspend withdrawal at will, it again weakens the commitment role of deposits. Historically, suspension was often collective and conditional, which may preserve some discipline while addressing systemwide shocks.

4.5 Relation to the literature

- **Diamond and Dybvig (1983).** Diamond and Rajan provide a foundation for why the assets funded by banks are illiquid in the first place. Banks do not merely repackage exogenous illiquidity; they enhance pledgeability.
- **Holmström and Tirole.** In their framework, intermediaries use collateral or pledges to support funding. Here, the bank improves financing because it can extract more from borrowers and commit to passing that value through.
- **Calomiris and Kahn (1991).** Calomiris and Kahn emphasize short-term debt as a monitoring device. Diamond and Rajan instead emphasize ex post discipline and disintermediation: depositors need not detect problems early, only be able to punish once problems arise.

4.6 Limitations and scope

- The main model abstracts from aggregate uncertainty and bank capital choice.
- Agents are risk neutral.
- The entrepreneur has all the bargaining power in renegotiation, although the authors argue the qualitative logic survives when the lender has some bargaining power too.
- The model is deliberately stripped down: the point is to isolate the commitment role of fragility.

5 Tables and figures

5.1 Figures 1 and 2: bargaining trees at dates 2 and 1

Read:

- Figure 1 shows the date 2 bargaining problem between the relationship lender and the entrepreneur. If the lender rejects renegotiation, she liquidates and gets X_2 .
- Figure 2 shows the date 1 bargaining problem when the lender is patient. The lender can accept, liquidate for X_1 , or continue toward date 2.
- Together, the two figures make clear why the entrepreneur's limited commitment constrains what can actually be promised in a financial contract.

Entrepreneur offers an alternative current payment, $P_2' < P_2$. He will supply current human capital and make the alternative current payment if and only if agreement is reached.

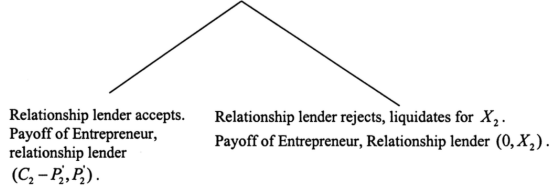


FIG. 1.—Bargaining at date 2 between a relationship lender and an entrepreneur

Entrepreneur offers an alternative current payment P_1' and date 2 renegotiation-proof payment P_2' . He will supply current human capital and make the alternative current payment if and only if agreement is reached.

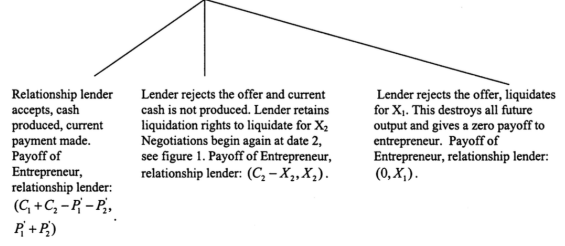


FIG. 2.—Bargaining at date 1 between a patient relationship lender and an entrepreneur

TABLE 1
TIME LINE

Date 0	Date 1	Date 2
1. Entrepreneur offers loan terms to the lender	1. Lender suffers (or does not suffer) a private liquidity shock	1. Entrepreneur makes any required payment or offers alternative schedule
2. Loan is made if the lender accepts	2. Entrepreneur makes any required payment or offers alternative schedule	2. If alternative schedule is offered, lender accepts or takes possession of asset
3. Amount is invested	3. If alternative schedule is offered, lender accepts or takes possession of asset	
4. Lender acquires specific expertise in liquidation through the relationship	4. Loan may be sold to another lender	

5.2 Table 1: timeline

Read:

- The table summarizes the basic chronology: loan contracting and investment at date 0, possible lender liquidity shock and renegotiation or sale at date 1, and final repayment or liquidation at date 2.
- It clarifies that the lender's relationship-specific expertise is acquired after the original loan is made.

5.3 Lemma 1 and the example: Figures 3 and 4

Read:

- Figure 3 plots the promised date-2 payment to the patient lender and the entrepreneur's net payoff in the numerical example.
- As the probability of a lender liquidity shock rises, the entrepreneur must promise more to the patient lender, so his net payoff falls.
- Figure 4 plots the illiquidity premium. It rises with the lender's liquidity risk, drops discretely when liquidation becomes part of the optimal contract, and eventually lending becomes infeasible.
- These figures are useful because they show that illiquidity is not just a conceptual problem — it leads to large changes in contract form and, eventually, the shutdown of direct finance.

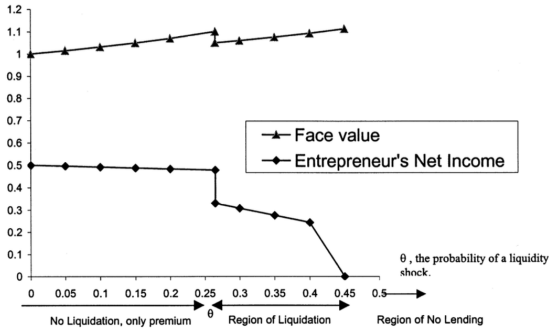


FIG. 3.—Date 2 face value when patient and entrepreneur's net income. The face value is V_2' (see the proof of lemma 2), and the entrepreneur's net income is given by the objective function in the entrepreneur's maximization problem (see the proof of lemma 2). Both values are in units of output.

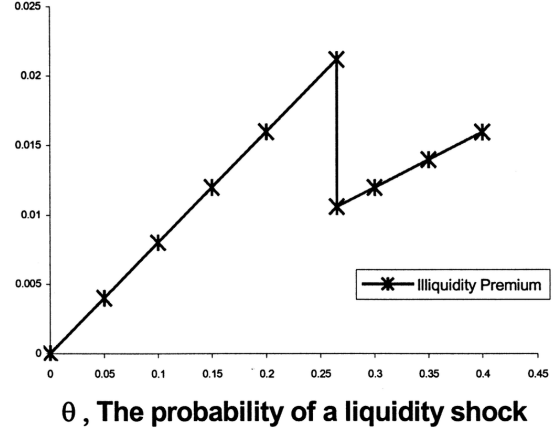


FIG. 4.—Illiquidity premium. The illiquidity premium is defined in the text and is in units of output (not percent).

5.4 Figure 5: depositor response to a low offer

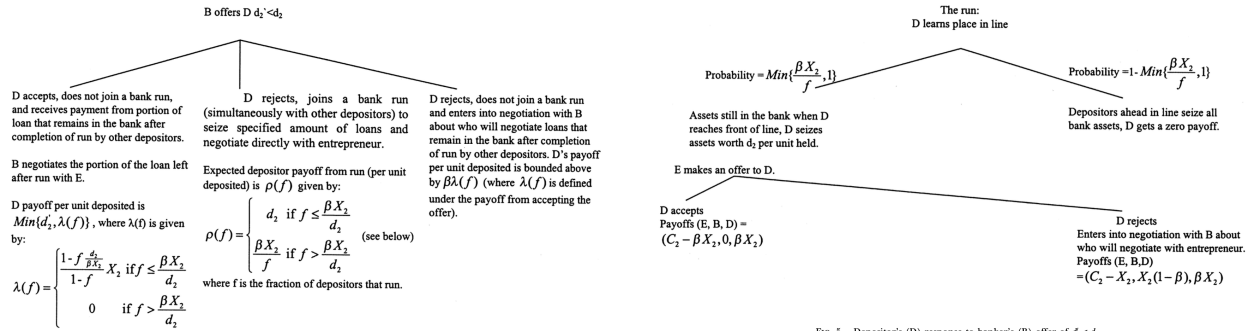


FIG. 5.—Depositor's (D) response to banker's (B) offer of $d_1' < d_2$

Read:

- Figure 5 illustrates the bank-run mechanism. If the banker offers depositors less than full value, each depositor has an incentive to demand payment first.
- The figure is the visual representation of the central commitment logic in the paper: the threat of a run prevents the banker from trying to keep rents for herself.

6 Short summary

- Method.** The paper builds a simple three-date theory of lending, illiquidity, and banking with relationship-specific collection skills, liquidity shocks, and renegotiation.
- Mechanism.** Direct relationship loans are illiquid because the lender cannot sell or pledge them for the full amount she can personally extract. Demand deposits create a credible commitment not to renegotiate with depositors.
- Main result.** A fragile bank capital structure can eliminate the illiquidity premium on loans and protect borrowers from lenders' liquidity needs, while simultaneously giving depositors liquid claims.

- **Implication.** Policies that make banks safer in a narrow sense may also reduce the liquidity creation function that makes banks valuable in the first place.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper shows that bank fragility can be socially useful because it helps a relationship lender commit to pass through the value of illiquid loans.
- In this way, demand deposits are not merely a financing choice layered on top of bank lending. They are part of the technology that makes bank lending possible.
- The model therefore unifies the asset side and liability side of banking: illiquid relationship loans and fragile liquid deposits are complementary.

7.2 Economic intuition

- A relationship lender can collect more from a borrower than outsiders can, but that very advantage makes the loan hard to sell.
- A bank solves this by making the lender vulnerable to a run. If she tries to hold up depositors, she loses control over the loan and therefore loses her rents.
- Fragility is useful because it threatens the one thing the banker values: continued control over the relationship asset.

7.3 Takeaway

This paper is important because it changes how we think about banks. Instead of seeing fragility only as a pathology, Diamond and Rajan show that fragility can be the commitment device that allows banks to create liquidity. The central message is subtle but powerful: making a financial intermediary *run-prone* can be exactly what lets it promise liquid claims against illiquid assets.